Partial Out-of-Distribution (POOD): Test queries include compositions involving both seen and unseen basic transformations, e.g., $f_1 \circ f_1 \to f_1 \circ f_2$. (*iv*) Out-of-Distribution (OOD). The test set contains entirely novel transformations (compositions) in training, e.g., $f_1 \circ f_1 \to f_2 \circ f_2$. The illustrative examples for transformation generalization under different scenarios are provided in Appendix C.1.1.

Table 1 | Full chain evaluation under different scenarios on transformation generalization.

Scenarios	Exact Match (%)	Edit Distance	BLEU Score
ID	100.00	0	1
CMP	0.01	0.1326	0.6867
POOD	0.00	0.1671	0.4538
OOD	0.00	0.2997	0.2947

Findings. Figure 3 illustrates the performance of the full chain under different distribution discrepancies. We can observe that, in general, the effectiveness of CoT reasoning decreases as the distribution discrepancy increases, which directly validates the data distribution lens. As shown in Table 1, CoT reasoning achieves satisfactory performance in the ID (exact match: 100%) scenario, while it degrades in CMP (0.01%), POOD (0%), and OOD (0%) scenarios. Diving into fine-grained analysis, as demonstrated in Table 2, we find that the success of CoT reasoning is attributed to the replicating pattern in the training data, as indicated by the inconsistency in reasoning and answers. For instance, when an unseen transformation $f_1 \circ f_1$ is present, LLMs attempt to generalize based on the most similar transformation (i.e., $f_1 \circ f_2$) seen during training, which leads to correct reasoning paths yet incorrect answers. As the commutativity of the transforms, generalization from $f_1 \circ f_2$ to $f_2 \circ f_1$ or vice versa allows LLMs to produce incorrect paths yet correct answers, which reflect the unfaithfulness and pattern-matching nature of CoT reasoning. Additional analysis and illustrative examples are provided in Appendix F.1 and G.1.

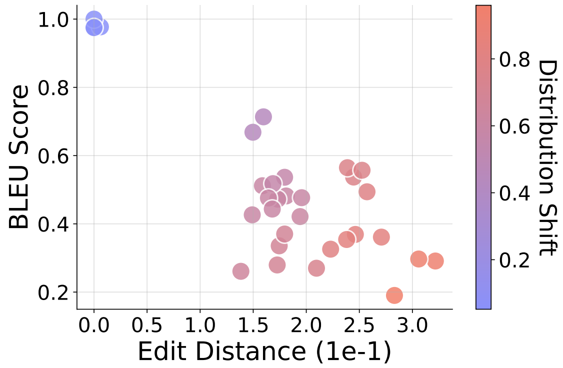

Figure 3 | Transformation generalization under different distribution discrepancies. The efficacy of CoT reasoning decreases as task distribution discrepancy increases.

Table 2 | Fine-grained analysis for CoT reasoning on transformation generalization based on exact match.

Transformation (Train → Test)	Reasoning	Answer	Full Chain
$\{f_1 \circ f_1, f_1 \circ f_2, f_2 \circ f_1\} \to f_2 \circ f_2$	100.00	0.01	0.01
$\{f_1 \circ f_2, f_2 \circ f_1, f_2 \circ f_2\} \to f_1 \circ f_1$	100.00	0.01	0.01
$f_1 \circ f_2 \to f_2 \circ f_1$	0.00	100.00	0.00
$f_2 \circ f_1 \to f_1 \circ f_2$	0.00	100.00	0.00

Experiment setup. To further probe *when* CoT reasoning can adapt to unseen transformations, we conduct supervised fine-tuning (SFT) experiments to incorporate a portion λ of unseen data.

Findings. As shown in Figure 4, we can find that generally a very small portion ($\lambda = 1.5e{-}4$) of data can make the model quickly generalize to unseen transformations. The less discrepancy between the training and testing data, the easier the model can generalize, highlighting the role of similar patterns that appear in the training data.

5.2. Element Generalization

Following a pipeline similar to transformation generalization, we investigate how CoT reasoning handles *elements* under various distribution discrepancies. Findings observed also support the proposed data distribution. The detailed experiment design and analysis can be found in Appendix F.2.

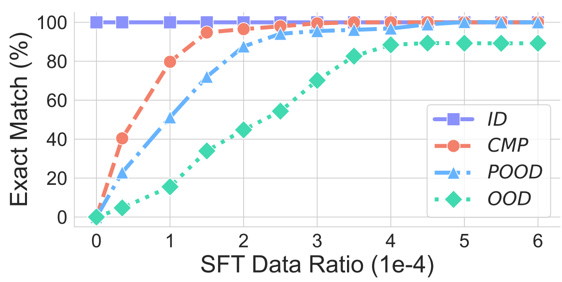

Figure 4 | Effectiveness of SFT. A small portion of unseen data helps CoT reasoning to quickly generalize.

6. Length Generalization

To study how CoT reasoning can operate on varying lengths, we design a length generalization experiment. Following the same intuition as *task generalization*, we also formulate length generalization as two perspectives: *text length* (i.e., element length) *generalization* and *reasoning step* (i.e., transformation length) *generalization*.

6.1. Text Length Generalization

Experiment setup. The text length distribution discrepancy can be measured by element length difference, detailed in Appendix E.2. We train LLMs on the dataset with text length $l = 4$ while fixing other factors and evaluate the performance on a variety of lengths (e.g., from $l = 2$ to $l = 6$). We provide illustrative examples for text length generalization in Appendix C.2.1.

Findings. As illustrated in the Figure 5, CoT reasoning produces excellent results under in-distribution scenarios ($l = 4$), while its performance degrades as discrepancies in the text length distribution increase, which serves as evidence for the data distribution lens. When we further analyze the exact match in the Table 5, the CoT reasoning failed to directly generate test cases for those lengths, even with a mild distribution shift (e.g., $l = 3$ or $l = 5$). Examples in Appendix G.2.2 indicate that LLMs attempt to produce CoT reasoning with the same length as the training data by adding or removing tokens when processing unseen text length. We further consider the effect of different padding strategies in Appendix F.3.

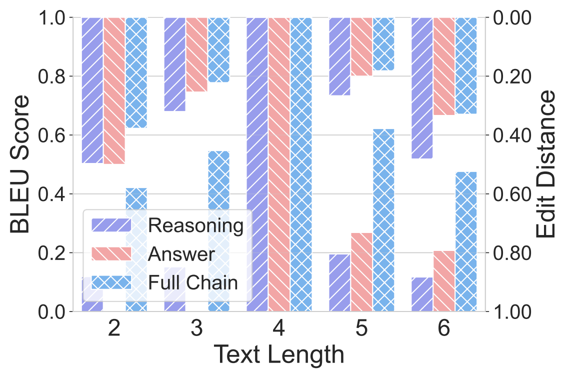

Figure 5 | Text length generalization under distribution discrepancies. Increasing distribution shifts in the text length lead to degraded CoT reasoning performance.

6.2. Reasoning Step Generalization

Experiment setup. Reasoning steps are determined by the number of basic *transformations* k in the *compositional transformation*. We mix the data with various reasoning steps (e.g., $k = 1, 2, 3$). By adjusting the mix ratio while maintaining the data size, we create different distribution discrepancies. Examples of reasoning step generalization are detailed in Appendix C.2.2.

Findings. As showcased in Figure 6, when we adjust the component of training data, the performance of CoT changed accordingly. For instance, increasing the ratio of $k = 1$ data will enhance performance on one-step reasoning while compromising two-step reasoning, which supports our hypothesis. Notably, considering extreme cases where the mix ratio is 0 or 1.0, the CoT reasoning achieves good

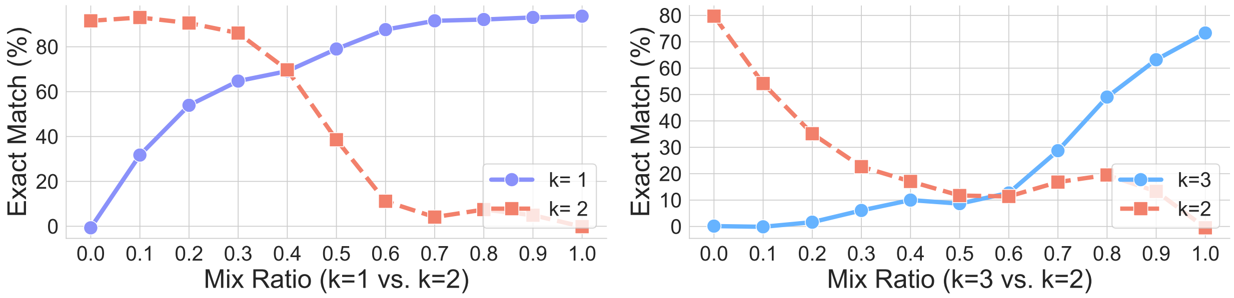

Figure 6 | Reasoning step generalization under distribution discrepancies. Performance of CoT reasoning systematically varies with training data components.

performance in the covered reasoning step but fails to generalize to unseen cases, indicating its fragility when encountering distribution shifts.

7. Format Generalization

To research the robustness of CoT reasoning when surface-level variations appear in test queries, we designed the *format generalization*.

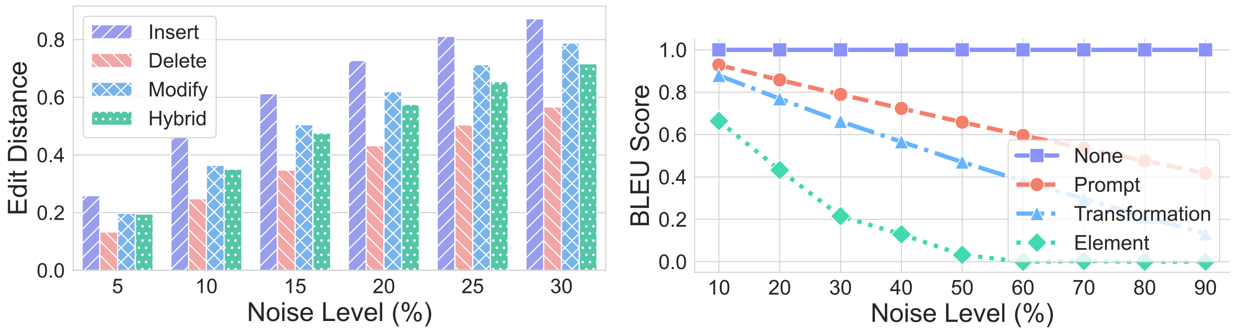

Figure 7 | Format generalization under distribution discrepancies. Testing performance degrades with various noise levels and in different applied areas.

Experiment setup. To introduce the distribution discrepancy at a formal level, we proposed distribution measurement (detailed in Appendix E.3) and consider four distinct perturbation modes to simulate a scenario in the real world. (i) Insert. One noise token is inserted; (ii) Delete. One original token is deleted; (iii) Modify: One original token is replaced with a noise token; and (iv) Hybrid: it combines the above-mentioned perturbation methods. We apply four perturbations with the noise level of p on different areas (e.g., elements, transformations, and prompts) of test queries. We further elaborate on the format generalization using illustrative examples in Appendix C.3.

Findings. As observed in Figure 7, introducing perturbation will compromise the effectiveness of CoT reasoning, and the degree depends on the noise level (i.e., distributional shift), which echoes the data distribution lens. Among different perturbation methods, insertion makes the greatest difference. Considering different areas applied, the elements and transformation play an important role, whereas the changes to other tokens have a lesser effect on the results, which aligns with intuition.